Revision Notes for Class 8 Science Chapter 8 – Nature of Matter Elements Compounds and Mixtures

Everything in our surroundings—the air, food, water, metals, and even the ground beneath us—is made up of matter. Matter consists of tiny particles and can be classified into mixtures and pure substances. Most items we use daily are not pure; they are combinations of two or more substances, making them mixtures. For instance, poha and sprout salad are mixtures of different visible ingredients, and solutions like lemonade and sugar water are examples where components blend so completely that they cannot be distinguished.

What Are Mixtures?

Mixtures are formed when two or more substances are combined in such a way that each keeps its own properties. The substances in a mixture are called its components. Mixtures can be non-uniform (heterogeneous), such as salads, where individual parts can be seen, or uniform (homogeneous), like salt dissolved in water, where the separate substances can't be seen even with a microscope. Air is another great example of a uniform mixture; it contains nitrogen, oxygen, argon, and carbon dioxide, all mixed so thoroughly that they cannot be easily separated by physical means.

Types of Mixtures

Mixtures are classified by the physical state of their components. For example, air (gas-gas) is uniform, while carbon particles in air (solid-gas) are non-uniform. Table 8.1 in the chapter shows different kinds of mixtures:

Mixture-type	Examples	Uniform/Non-uniform
Gas + Gas	Air	Uniform
Gas + Liquid	Soda water, oxygen in water	Uniform

Solid + Gas	Dust in air	Non-uniform
Liquid + Liquid	Vinegar in water, oil and water	Uniform (vinegar), Non-uniform (oil-water)
Solid + Liquid	Sand in water, seawater	Non-uniform (sand), Uniform (seawater)
Solid + Solid	Baking powder, alloys	Uniform (alloys), Non-uniform (baking powder)

Mixtures can be separated by physical means, such as filtration or evaporation, to obtain their pure substances.

What Are Pure Substances?

The word 'pure' in science is different from how we use it every day. A pure substance means it contains only one type of particle and cannot be separated into simpler substances through physical processes. Even if a food product says 'pure' on its label, in scientific terms, it may still be a mixture if it's made up of more than one substance. For example, milk and fruit juice, though considered pure in daily life, are actually mixtures in science.

Types of Pure Substances: Elements and Compounds

Pure substances are of two main types: elements and compounds. Elements are the simplest substances; they cannot be broken down further. Examples include gold, carbon, iron, and sulfur. An element consists of only one kind of atom. Many elements exist as molecules made up of the same kind of atom—for example, O_2 (oxygen).

Compounds are made when two or more different elements combine chemically in a fixed ratio. For example, water is always made from two hydrogen atoms and one oxygen atom (H_2O). Compounds have properties that are different from their constituent elements. Once formed, the elements in a compound cannot be separated by physical methods. Salt (sodium chloride) and sugar are common examples of compounds.

Experiment Examples: Separating Elements and Compounds

When electricity is passed through water, it splits into two gases: hydrogen (which explodes with a pop) and oxygen (which makes a flame glow brighter). This process shows that water is a compound composed of these elements. Heating sugar breaks it down into carbon (black residue) and water vapour, showing it's also a compound. On the other hand, a mixture of iron filings and sulfur can be separated by using a magnet, but heating this mixture forms iron sulfide—a compound with different properties that no longer attracts to a magnet and smells like rotten eggs when treated with acid.

Elements, Compounds, and Mixtures in Daily Use

Everything we use or see is either an element, compound, or mixture. Air, for example, is a mixture of gases; seawater contains a mix of water and salts. Elements like iron and aluminium are used to make cars, bridges, and buildings. Compounds such as medicines and fertilizers are essential for health and agriculture. Alloys like bronze and stainless steel, made by combining metals, are mixtures designed for specific purposes.

Minerals

Most rocks are mixtures of different minerals. Some minerals, called native minerals, are pure elements like gold, silver, or carbon (as graphite or diamond). However, most minerals are compounds, for instance quartz (silicon dioxide) or mica. Everyday items like cement and talcum powder are made from minerals.

Snapshots – Key Points

- A mixture consists of two or more substances that keep their own properties and can often be separated by physical means.
- Components of a mixture do not react chemically with each other.

- A pure substance contains only one kind of particle and cannot be separated into different substances by physical processes.
- Elements are made of only one kind of atom and can't be broken down further.
- Compounds are formed by chemical combination of elements in fixed ratios and have properties different from the individual elements.
- Minerals are natural solid substances, mostly compounds, but some are pure elements.
- Physical methods can separate mixtures, but not elements or compounds.

Traditional and Modern Applications

In ancient India, alloys like bronze (called Kamsya, a mix of copper and tin) were used for making utensils and for medicinal purposes. Today, combinations of elements and compounds are used to create modern building materials, medicines, fertilizers, and even advanced materials like graphene aerogel, which is the world's lightest solid and useful for cleaning oil spills.

Understanding the differences between mixtures, elements, and compounds helps us appreciate how we use resources and how scientists and engineers create new materials to solve everyday challenges.